

# Лабораторная работа № 4

Эмуляция и измерение задержек в глобальных сетях

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Основной целью работы является знакомство с NETEM — инструментом для тестирования производительности приложений в виртуальной сети, а также получение навыков проведения интерактивного и воспроизводимого экспериментов по измерению задержки и её дрожания (jitter) в моделируемой сети в среде Mininet.

```
mininet@mininet-vm:~$ xauth list $DISPLAY
mininet-vm/unix:11 MIT-MAGIC-COOKIE-1 f0cb9ac988b1dac1b6396527ccbb5060
mininet@mininet-vm:~$ sudo -i
root@mininet-vm:~# xauth add mininet-vm/unix:11 MIT-MAGIC-COOKIE-1 f0cb9ac988b1dac1b6396527ccbb5060
root@mininet-vm:~# logout
mininet@mininet-vm:~$
```

Рис. 1: Исправление прав запуска X-соединения

Простейшая топология

```
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Running terms on localhost:11.0
*** Starting controller
c0 X11 connection rejected because of wrong authentication.

X11 connection rejected because of wrong authentication.
*** Starting 1 switches
s1 ...X11 connection rejected because of wrong authentication.
X11 connection rejected because of wrong authentication.

*** Starting CLI:
```

Рис. 2: ifconfig на хостах h1 и h2

## Выполнение лабораторной работы

```
root@mininet-vn:/home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether ee:cf:43:e2:c4:1a txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1646 bytes 359400 (359.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1646 bytes 359400 (359.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vn:/home/mininet#
```

```
root@mininet-vn:/home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether ae:8b:31:4f:fe:28 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 2198 bytes 389792 (389.7 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2198 bytes 389792 (389.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vn:/home/mininet#
```

# Выполнение лабораторной работы

```
root@h1:~# ifconfig
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1646 bytes 359400 (359.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1646 bytes 359400 (359.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping -c 6 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.207 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.047 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 508ms
rtt min/avg/max/mdev = 0.047/1.861/10.741/3.971 ms
root@mininet-vm:/home/mininet#
```

```
root@h2:~# ifconfig
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 2198 bytes 389792 (389.7 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2198 bytes 389792 (389.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-vm:/home/mininet# ping -c 6 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.51 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.048 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.050 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.042 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.045 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.048 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 508ms
rtt min/avg/max/mdev = 0.042/0.456/2.505/0.916 ms
root@mininet-vm:/home/mininet#
```



```
x "host: h1"@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.207 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.056 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.047 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5086ms
rtt min/avg/max/mdev = 0.047/1.861/10.741/3.971 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 100ms
root@mininet-vm:/home/mininet# ping -c 6 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=101 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 501ms
rtt min/avg/max/mdev = 100.821/101.128/102.498/0.631 ms
root@mininet-vm:/home/mininet#
```

Рис. 5: Двухнаправленная задержка соединения

```
root@mininet-vm:~# host: h1@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=10.7 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.207 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.056 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.047 ms
:
--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5086ms
rtt min/avg/max/mdev = 0.047/1.861/10.741/3.971 ms
root@mininet-vm:~/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 100ms
root@mininet-vm:~/home/mininet# ping -c 6 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=101 ms
--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5011ms
rtt min/avg/max/mdev = 100.621/101.128/102.498/0.631 ms
root@mininet-vm:~/home/mininet# sudo tc qdisc change dev h1-eth0 root netem delay 100ms
```

Рис. 6: Изменение задержки на 50мс

# Выполнение лабораторной работы

```
root@host h1@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=101 ms
:
--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 501ms
rtt min/avg/max/ndev = 100,621/101,128/102,498/0,631 ms
root@mininet-vm:/home/mininet# sudo tc qdisc change dev hl-eth0 root netem delay 50ms
root@mininet-vm:/home/mininet# ping -c 6 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=103 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=103 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=101 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 501ms
rtt min/avg/max/ndev = 100,687/101,868/103,474/1,002 ms
root@mininet-vm:/home/mininet#

root@host h2@mininet-vm

root@mininet-vm:/home/mininet# ping -c 6 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=2.51 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.048 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.050 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.042 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.045 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.048 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 508ms
rtt min/avg/max/ndev = 0,042/0,456/2,505/0,915 ms
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h2-eth0 root netem delay 100ms
root@mininet-vm:/home/mininet# ping -c 6 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=203 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=201 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=201 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=201 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=201 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=201 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 501ms
rtt min/avg/max/ndev = 200,647/201,308/202,957/0,760 ms
root@mininet-vm:/home/mininet# sudo tc qdisc change dev h2-eth0 root netem delay 50ms
root@mininet-vm:/home/mininet# ping -c 6 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=102 ms
```

# Выполнение лабораторной работы

```
host: h1" @mininet-vm

64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=103 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=103 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=101 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5013ms
rtt min/avg/max/ndev = 100,887/101,888/103,474/1,002 ms
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# ping -c 6 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2,50 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0,043 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0,120 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0,044 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0,049 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0,075 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5079ms
rtt min/avg/max/ndev = 0,043/0,471/2,500/0,507 ms
root@mininet-vm:/home/mininet#

host: h2" @mininet-vm

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5011ms
rtt min/avg/max/ndev = 200,647/201,308/202,367/0,760 ms
root@mininet-vm:/home/mininet# sudo tc qdisc change dev h2-eth0 root netem delay 50ms
root@mininet-vm:/home/mininet# ping -c 6 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=102 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=101 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=102 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=102 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=102 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=101 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5011ms
rtt min/avg/max/ndev = 100,829/101,735/102,331/0,496 ms
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h2-eth0 root netem
root@mininet-vm:/home/mininet# ping -c 6 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=3,72 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0,435 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0,172 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0,049 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0,061 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0,051 ms

--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5077ms
rtt min/avg/max/ndev = 0,049/0,757/3,721/1,334 ms
```

```
root@mininet-vn:~# "host: h1"@mininet-vn

64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.120 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.044 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.075 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5079ms
rtt min/avg/max/mdev = 0.043/0.471/2.500/0.907 ms
root@mininet-vn:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
Error: Invalid qdisc name.
root@mininet-vn:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 100ms 10ms
root@mininet-vn:/home/mininet# ping -c 6 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=111 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=95.8 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=100 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=95.5 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=92.3 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=93.9 ms

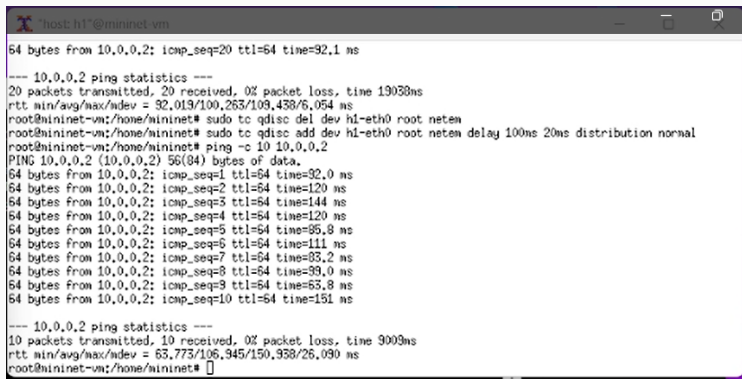
--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5011ms
rtt min/avg/max/mdev = 92.281/98.129/111.010/6.262 ms
root@mininet-vn:/home/mininet#
```

Рис. 9: Добавление значения корреляции для джиттера и задержки в интерфейс подключения

```
root@mininet-vm:~# "host: h1"@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=93.9 ms

--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 501ms
rtt min/avg/max/mdev = 92.281/98.129/111.010/6.262 ms
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 100ms 10ms 25%
root@mininet-vm:/home/mininet# ping -c 20 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=93.1 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=108 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=102 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=94.1 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=109 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=109 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=99.1 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=92.0 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=107 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=92.1 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=101 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=96.9 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=96.3 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=96.6 ms
```

Рис. 10: Распределение задержки в интерфейсе подключения

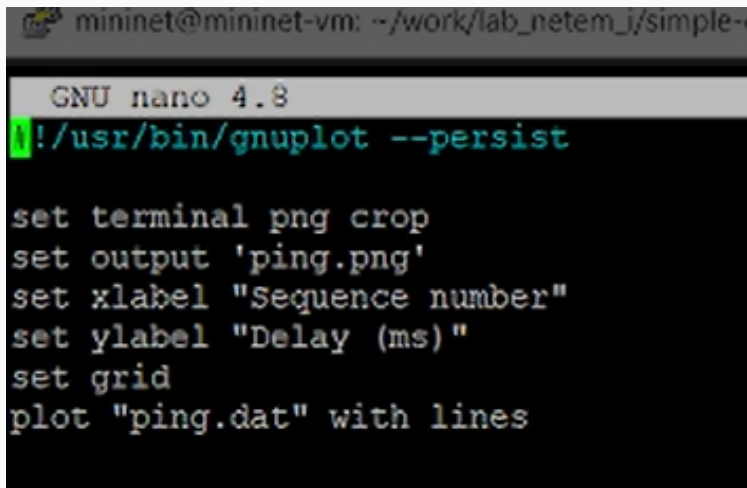


```
host: h1*@mininet-vm
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=92.1 ns

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19038ms
rtt min/avg/max/mdev = 92.019/100.263/109.438/6.054 ms
root@mininet-vm:/home/mininet# sudo tc qdisc del dev h1-eth0 root netem
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 100ms 20ms distribution normal
root@mininet-vm:/home/mininet# ping -c 10 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=92.0 ns
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=120 ns
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=144 ns
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=120 ns
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=85.8 ns
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=111 ns
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=83.2 ns
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=99.0 ns
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=63.8 ns
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=151 ns

--- 10.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9009ms
rtt min/avg/max/mdev = 63.773/106.945/150.938/26.090 ms
root@mininet-vm:/home/mininet#
```

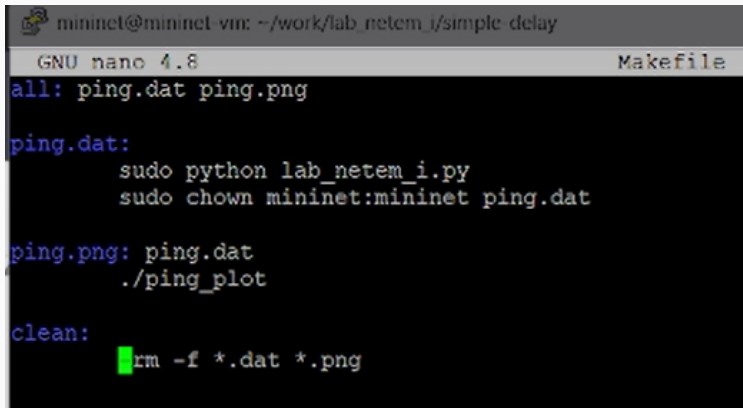
Рис. 11: скрипт для визуализации ping\_plot



```
mininet@mininet-vm: ~/work/lab_netem_j/simple-  
GNU nano 4.8  
#!/usr/bin/gnuplot --persist  
  
set terminal png crop  
set output 'ping.png'  
set xlabel "Sequence number"  
set ylabel "Delay (ms)"  
set grid  
plot "ping.dat" with lines
```

Рис. 12: Makefile для управления процессом проведения эксперимента



A screenshot of a terminal window showing a nano editor editing a Makefile. The terminal title bar indicates the user is 'mininet' on a 'mininet-vm' machine, in the directory '~/work/lab\_netem\_i/simple\_delay'. The nano editor's status bar shows 'GNU nano 4.8' and 'Makefile'. The Makefile content is as follows:

```
all: ping.dat ping.png

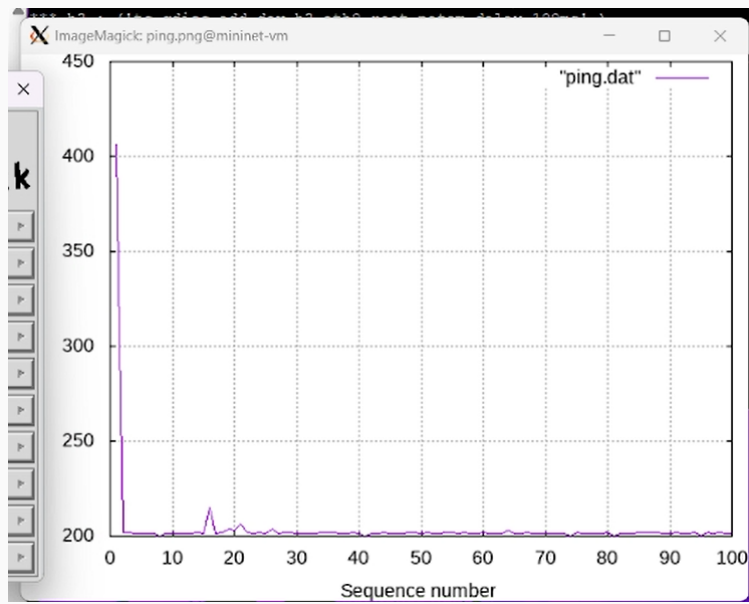
ping.dat:
    sudo python lab_netem_i.py
    sudo chown mininet:mininet ping.dat

ping.png: ping.dat
    ./ping_plot

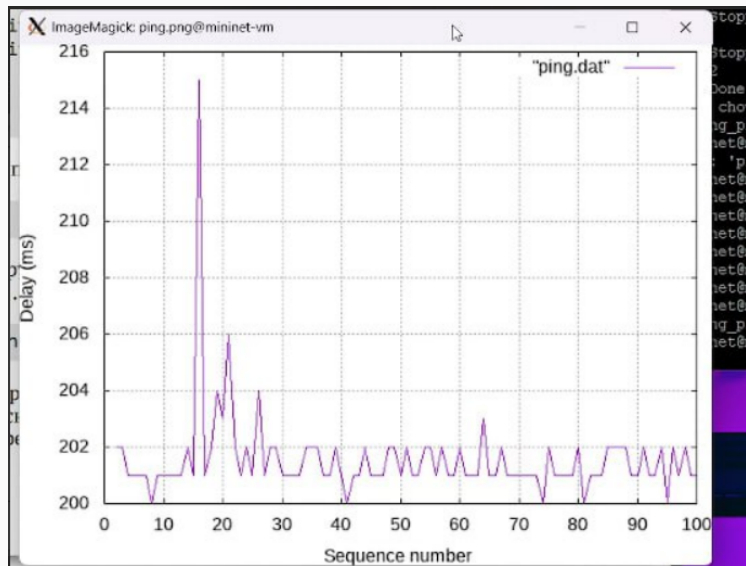
clean:
    rm -f *.dat *.png
```

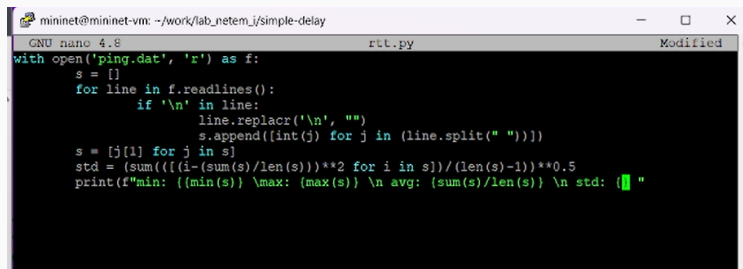
Рис. 13: результате выполнения скриптов

## Выполнение лабораторной работы



## Выполнение лабораторной работы





```
mininet@mininet-vm: ~/work/lab_netem_j/simple-delay
GNU nano 4.8      rtt.py      Modified
with open('ping.dat', 'r') as f:
    s = []
    for line in f.readlines():
        if '\n' in line:
            line.replacr('\n', "")
            s.append([int(j) for j in (line.split(" "))])
s = [j[1] for j in s]
std = (sum([(i-(sum(s)/len(s)))**2 for i in s])/(len(s)-1))**0.5
print(f"min: {(min(s)} \max: {max(s)} \n avg: {sum(s)/len(s)} \n std: { "

```

Рис. 16: Результат работы скрипта rtt.py

В результате выполнения данной лабораторной работы я познакомилась с NETEM – инструментом для тестирования производительности приложений в виртуальной сети, а также получила навыки проведения интерактивного и воспроизводимого экспериментов по измерению задержки и её дрожания (jitter) в моделируемой сети в среде Mininet.